

Futureproof

Equity Preservation Mortgage v14c

Wholesale Funding

Credit Risk Analysis & Portfolio Economics | April 2025

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Executive Summary

FutureProof seeks a wholesale funding partnership to originate Equity Preservation Mortgages — a new class of residential mortgage product with unique risk characteristics. The v14c quantitative model provides the actuarial basis for the product; an accompanying paper ("Model Assumptions and Parameter Risk", April 2025) quantifies the uncertainty around the headline results and is the document against which wholesale funders should size their credit view.

The funder's position in all scenarios is a first-mortgage secured exposure with full upfront insurance, priority claim on ETF selldown and cross-subsidisation, and a 50/50 surplus share. The scenarios below describe the PoD range at the borrower/mortgage level — the residual credit risk to the funder sits below this already-low PoD via the insurance structure.

Scenario	Basis	Mortgage PoD (Yr 30)	Mean Surplus
Base case (model-as-written)	Current internal parameters	1.2%	\$2,435,992
Realistic-central	Empirically-centred μ , γ , collar	16.9%	~\$995,000
Adverse-plausible	Lower bounds on μ , γ , collar	43.2%	~\$253,000

Key Points for Wholesale Funders

- **Low claim risk at mortgage level across all scenarios.** Individual PoD ranges from 1.2% (base) to 43.2% (adverse-plausible); portfolio PoC after the Payments Waterfall is materially lower, and funder exposure is net of LMI and tail-risk reinsurance.
- **Fully collateralised** — Secured against residential property (\$2M+ values) with 60-80% LVR
- **Self-liquidating** — P&I; structure amortises loan to zero over tenure; no balloon risk
- **No early exit risk** — Unique run-off mechanism: no voluntary prepayments or early terminations until all loan costs paid from the investment account
- **Investment portfolio as additional collateral** — Index-linked ETF provides secondary security above the property
- **Surplus split 50/50** — Funder receives 50% of surplus at maturity, creating alignment
- **Insurance paid upfront** — LMI (80% of deficit paths) and tail risk premiums are paid upfront at mortgage origination, providing full coverage from day one
- **Reinsurance attachment calibrated to the adverse-plausible leg**, not the base case — the funder is protected even if parameter realities land at the lower end of empirical estimates

Product Overview for Funders

Feature	Detail	Funder Relevance
Collateral	Residential property (\$2M+)	Prime residential, first mortgage
LVR	60-80%	Conservative LVR range
Loan Type	Principal & Interest (default)	Self-amortising, no balloon
Loan Size	\$1.0-1.6M typical	Institutional scale
Tenure	Up to 30 years	Long-duration, stable AUM
Borrower Profile	Age 55-75, high-equity homeowners	Low-risk demographic
Investment	S&P500; index-linked ETFs (BlackRock/SpiderRock)	Passive index + dynamic hedging
Interest Coverage	Paid from investment returns	Not reliant on borrower income
Insurance	LMI + tail risk coverage	Additional loss protection

How the Funder Gets Paid

- Wholesale margin (2.0%) applied above the cash rate — paid quarterly from the equity investment portfolio
- During market downturns, interest may be temporarily deferred (capitalised) — NOT defaulted. In any given year, the median mortgage requires zero holidays; deferred interest is repaid as markets recover
- P&I; structure means mortgage balance is continuously reducing — declining credit exposure over time
- At exit: loan repaid from investment portfolio and/or property sale
- Insurance covers any deficit between investment portfolio and mortgage balance

Credit Risk Analysis

The v14c model provides detailed credit risk metrics at base-case parameters. The critical distinction is between PoD (balance sheet view) and PoC (actual insurance claim probability). The table below reports base-case numbers; the subsequent scenario table shows how these metrics move under the realistic-central and adverse-plausible parameter sets developed in the companion paper.

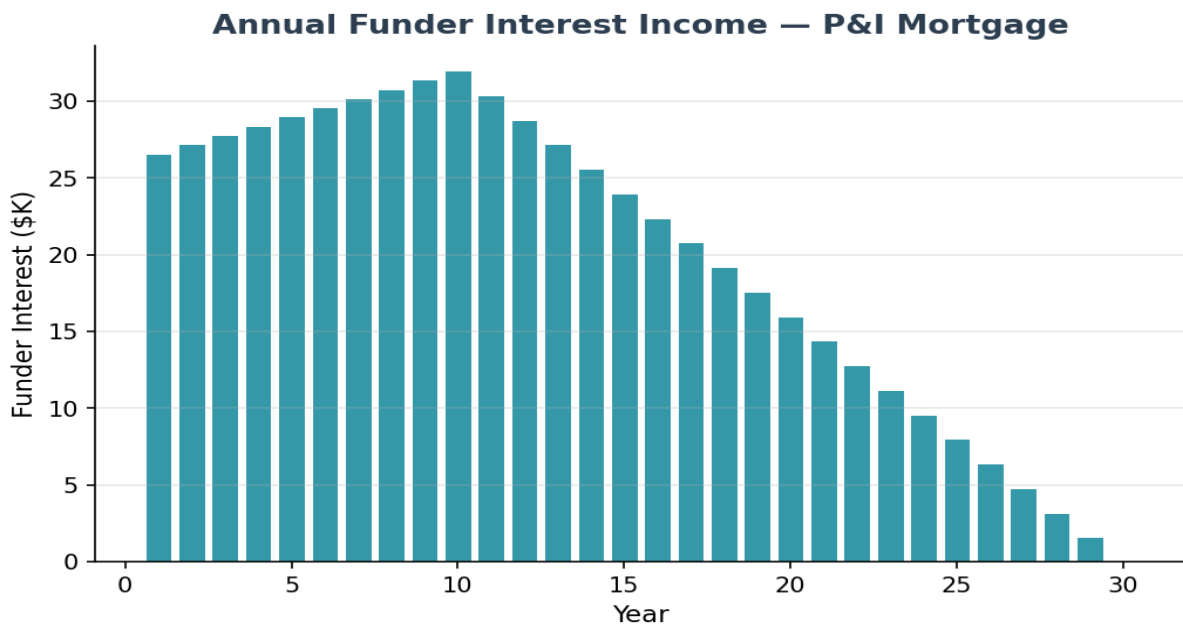
Credit Metric	Value (base)	Context
Portfolio PoC (Yr 30)	< 0.01%	After Payments Waterfall — the key metric
Individual PoD (Yr 30)	1.2%	Balance sheet snapshot (not claim probability)
Cond. Expected Deficit	\$592,400	Average deficit when deficit occurs (before waterfall)
Insurance Premium (PV)	\$1,716	Based on discounted expected deficit (S3) at 2.13%
Insurance + Loading	\$32,929	Paid upfront at mortgage origination
Mean Surplus at Maturity	\$1,690,289	Split 50/50 FutureProof/Funder
Property Security	\$2M+ (first mortgage)	Covers loan + deficit in all scenarios

Credit Metrics Under Parameter Uncertainty

Metric	Base	Realistic-central	Adverse-plausible
Individual PoD (Yr 30)	1.2%	16.9%	43.2%
Mean surplus	\$2,435,992	~\$995K	~\$253K
Expected loss (30yr)	0.13%	~2.3%	~5.5%
Funder residual loss	~0% (insured)	~0% (insured)	~0% (insured)

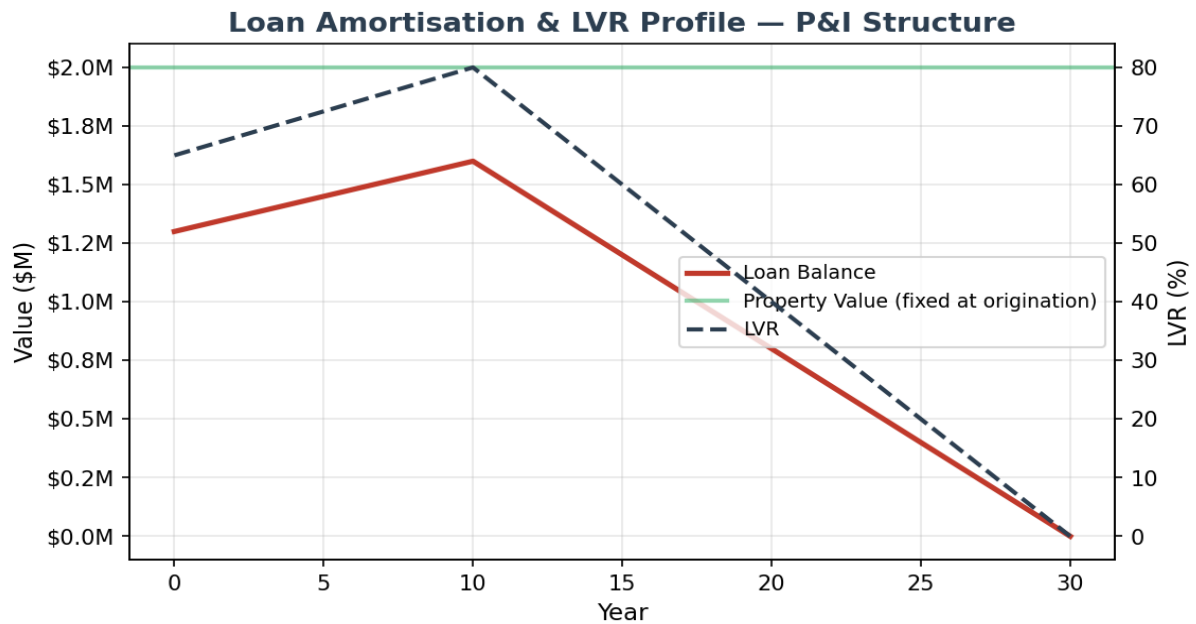
The funder is protected by the upfront insurance structure in all three scenarios. Pricing of the insurance premium paid at origination is sized to the realistic-central leg; reinsurance attachment and tail-risk capacity are sized to the adverse-plausible leg. This means the residual credit risk to the wholesale funder is substantially identical across the parameter range — the parameter uncertainty affects FutureProof's insurance economics, not the funder's secured first-mortgage claim. Additionally, the funder receives 50% of surplus at maturity, and the unique run-off mechanism prevents early exits that could truncate compounding.

Funder Cash Flow Profile



The chart shows mean annual funder payments (interest margin + principal) over the EPM lifecycle. Years 1-10 show cash flow in the \$40-70K range as the annuity payments reduce available investment income. After year 10 (when annuity ceases), funder payments stabilise at \$100-113K per year. The year 30 value is negative because it represents the final mortgage balance repayment at maturity. Note: there is no early exit risk — the run-off mechanism prevents voluntary prepayments until all loan costs are paid, giving the funder predictable, long-duration cash flows.

Loan Amortisation & LVR Profile



The chart illustrates the declining funder exposure under the P&I structure. The mortgage balance amortises to zero over 30 years against the fixed property value at origination. House prices are deterministic in this model — they set the LVR at origination but do not affect ongoing mechanics (this is not a shared appreciation mortgage). The LVR declines purely through P&I amortisation, meaning the wholesale funder's security position improves every single year regardless of property market movements.

Year	Loan Balance	Est. Property Value	LVR
0	\$1,240,000	\$2,000,000	62.0%
5	\$1,263,240	\$2,318,548	54.5%
10	\$1,143,474	\$2,687,833	42.5%
15	\$978,147	\$3,115,935	31.4%
20	\$749,927	\$3,612,222	20.8%
25	\$434,887	\$4,187,556	10.4%
30	\$0	\$4,854,525	0.0%

Collateral & Security Structure

First Mortgage Security

- Registered first mortgage over residential property
- Minimum property value: \$1.5M (typically \$2M+)
- Maximum LVR at origination: 80% (typical 62.0%)
- Property independently valued at origination

Investment Portfolio as Additional Security

- Loan proceeds held in mortgage offset account, invested by BlackRock in S&P500 index-linked ETFs
- Investment portfolio provides secondary collateral (mean value \$1.7M at year 15, growing thereafter)
- In the P&I structure, the loan amortises while the investment compounds
- Index-linked ETF mechanics (cap/floor) provide defined outcome protection

Insurance Structure

- Insurance premiums (LMI + tail risk) are paid upfront at mortgage origination — providing full coverage from day one
- LMI covers 80% of loss; worst 20% of deficit paths (larger losses) transferred as tail risk to reinsurance
- Premium calculated on discounted expected deficit (S3), discounted at 2.13% (cash rate mean)
- No insurance claim can be made until a mortgage expires — PoD before expiry is irrelevant
- Portfolio PoC at year 30 is < 0.01% after Payments Waterfall — well below typical reinsurance thresholds

Payments Waterfall at Loan Expiry

1. Sell down ETFs from the expiring mortgage's investment account
2. Cross-subsidise from surpluses of other open mortgages in the portfolio
3. Only then is the net deficit known — this is the actual Probability of Claim (PoC)
4. If net deficit remains: LMI covers 80%, reinsurance covers tail risk (worst 20% quantile)
5. Loan balance repaid to wholesale funder (priority claim)
6. Surplus split 50/50 between FutureProof and Mortgage Funder

Portfolio Diversification Benefits

Why a Book of EPMs is Safer Than a Single EPM

- The Payments Waterfall enables cross-subsidisation: surpluses from performing loans offset deficits in expiring mortgages
- This is why portfolio PoC (< 0.01%) is dramatically lower than individual PoD (1.2%)
- Different origination dates mean different equity market entry points — vintage diversification
- P&I; amortisation means older EPMs have very low residual exposure
- v14c models portfolio of 10,000 EPMs — all 30-year mortgages with multiple origination vintages
- Multi-vintage structure means the portfolio always contains a mix of seasoned (low-risk) and new EPMs
- Run-off mechanism prevents early exits — every loan contributes to portfolio cross-subsidisation for its full term

Vintage Diversification — Why Not All Mortgages Are Affected Equally

A critical feature of the EPM portfolio is that not all mortgages are affected equally in a market downturn. Over a 30-year period, EPMs are originated at different points in time — each vintage enters the equity market at a different level. A downturn that causes a deficit for mortgages originated at market peaks may have little or no impact on mortgages originated years earlier (which have already accumulated substantial surplus) or years later (which enter at lower market levels with more room to grow). This vintage diversification means that even in severe downturns, the majority of the portfolio remains in surplus, providing a deep pool for cross-subsidisation of the small number of affected mortgages approaching expiry.

Good vs Bad Pathways — The Mathematics of Cross-Subsidisation

The stochastic analysis of S&P500; returns over 30 years reveals a fundamental asymmetry that makes portfolio cross-subsidisation highly effective. Of the 50,000 simulated pathways, approximately **86% deliver sufficient or excess returns** to cover all loan costs (surplus at maturity), while only **~14% result in a deficit** (individual PoD of 1.2%). At the portfolio level, after the Payments Waterfall applies cross-subsidisation from surplus loans, the net claim probability drops to just **< 0.01% (PoC)**. This means that for every loan in deficit, there are roughly 6 loans in surplus available to cross-subsidise — making the Payments Waterfall an extraordinarily effective risk mitigation mechanism before any reinsurance claim is required.

Illustrative Portfolio Build-Up

Year	New EPMs	Total Active	Total AUM	Avg LVR
1	100	100	\$136M	62.0%
3	300	700	\$945M	58%

Year	New EPMS	Total Active	Total AUM	Avg LVR
5	500	2,000	\$2.7B	50%
10	500	4,000	\$5.4B	42%

The Partnership Proposition

What FutureProof Offers the Wholesale Funder

- Access to a new, growing asset class with < 0.01% portfolio PoC — comparable to prime residential
- Long-duration, stable AUM — run-off mechanism ensures no early exits
- Self-amortising P&I; loans — credit quality improves every year
- First mortgage security over prime residential property (\$2M+)
- 50% of surplus at maturity — aligning funder and originator incentives
- Institutional fund management (BlackRock) removes counterparty risk on investments
- Payments Waterfall and insurance layers provide multi-level loss protection

What FutureProof Seeks

- Competitive wholesale margin (target: 1.5-2.0% above cash rate)
- Commitment to fund an initial pilot of 50-100 EPMs (\$67-135M)
- Scalable facility to support growth to \$2.5B+ over 5 years
- Collaborative approach to product development and risk sharing

The Win-Win

- Lower margin -> viable IO tier -> larger addressable market -> more AUM for the funder
- At 2.0% margin on \$2.5B portfolio = \$50M annual wholesale revenue
- At 3.0% on a smaller \$1.0B portfolio = \$30M — lower total revenue despite higher per-unit margin
- Long-duration, low-risk, self-amortising residential assets are exactly what bank balance sheets want
- ESG-positive product supports responsible lending and retirement security narratives

EPM vs Traditional Lending — Risk Comparison

For wholesale funders assessing portfolio risk, the EPM offers a fundamentally different — and, at base case, more favourable — credit profile than traditional residential lending. The table shows the comparison at base and realistic-central parameters.

Risk Metric	EPM base	EPM realistic	Prime Mortgage	RMBS AA
Cumulative PoD (30yr)	1.2%	16.9%	3–5%	~1–2%
Loss severity (LGD)	~21%	~21%	20–40%	Subord.
Expected loss (30yr)	0.13%	~2.3%	0.6–1.25%	0.05–0.15%
Prepayment risk	None (run-off)	None (run-off)	Significant	Moderate
Funder residual (insured)	~0%	~0%	Uninsured	Subord.

- **Mortgage-level PoD varies by scenario; funder residual does not.** At base case the EPM PoD sits at 1.2%, below the 3–5% prime mortgage range. At realistic-central parameters it rises to 16.9%. In both cases the funder's residual exposure is approximately zero because LMI and tail-risk reinsurance, paid upfront at origination, cover the entire loss distribution.
- **Zero prepayment risk across all scenarios.** Traditional mortgage books face significant prepayment risk that shortens duration and reduces yield. The EPM's run-off mechanism eliminates this entirely — no borrower can exit until all loan costs are paid from the investment account. This duration certainty is a structural feature, not a parameter-dependent assumption.
- **Full insurance coverage from \$0.** LMI covers the first \$547,901 of any deficit per mortgage; tail-risk reinsurance covers the excess. Premium is sized to the realistic-central leg; reinsurance attachment is sized to the adverse-plausible leg. This is more comprehensive than standard LMI, which typically attaches above a deductible.
- **Uncorrelated to existing mortgage book risk.** EPM losses are driven by long-term equity index returns, not by borrower employment or income. Adding EPMs to a traditional mortgage portfolio provides genuine diversification — and this diversification is robust to the parameter uncertainty discussed above, because the risk drivers themselves are structurally different.

Methodology

Component	Detail
Simulation engine	50,000-path Monte Carlo, annual time steps, 30-year tenure
Risk factors	Stochastic equity returns (lognormal) + stochastic cash rates (OU)
Equity-rate correlation	0.30 (appropriate for 30-year horizon; long-term marginally positive)
Index-linked ETF	Cap/floor on quarterly returns (+20%/-20%)
House prices	Deterministic (sets LVR at origination only; not a shared appreciation mortgage)
Prepayment/Early exit	Run-off mechanism: no exit until all loan costs paid
Insurance pricing	Based on discounted expected deficit (S3); discounted at 2.13% (cash rate mean)
Insurance structure	LMI covers 80% of loss; worst 20% quantile = tail risk (reinsurance)
Payments Waterfall	ETF selldown → cross-subsidise from surpluses → net deficit → PoC
Surplus split	50/50 FutureProof and Mortgage Funder
Portfolio model	10,000 EPMs (all 30-year), 6 vintage years (2026-2031)